



Top 17 CAT Coordinate Geometry Questions With Video Solutions

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Questions

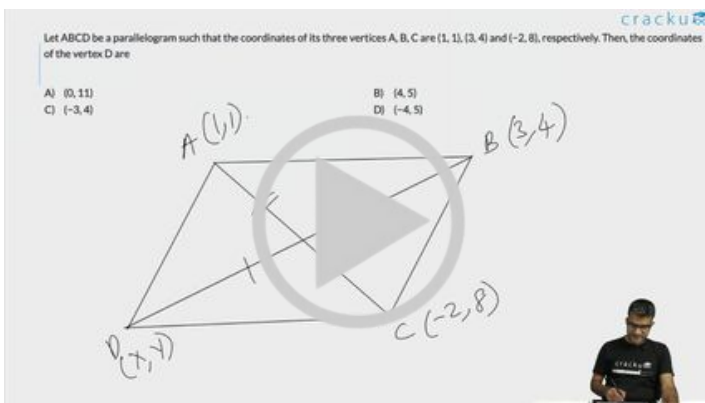
Instructions

For the following questions answer them individually

Question 1

Let ABCD be a parallelogram such that the coordinates of its three vertices A, B, C are (1, 1), (3, 4) and (-2, 8), respectively. Then, the coordinates of the vertex D are

- A (0, 11)
- B (4, 5)
- C (-3, 4)
- D (-4, 5)



VIDEO SOLUTION

Question 2

The points (2,1) and (-3,-4) are opposite vertices of a parallelogram. If the other two vertices lie on the line $x + 9y + c = 0$, then c is

- A 12
- B 13
- C 15
- D 14

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The points $(2,1)$ and $(-3,-4)$ are opposite vertices of a parallelogram. If the other two vertices lie on the line $x + 9y + c = 0$, then c is

A) 12 B) 13
C) 15 D) 14

VIDEO SOLUTION

Question 3

The vertices of a triangle are $(0,0)$, $(4,0)$ and $(3,9)$. The area of the circle passing through these three points is

- A $\frac{14\pi}{3}$
B $\frac{123\pi}{7}$
C $\frac{12\pi}{5}$
D $\frac{205\pi}{9}$

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C) $\frac{12\pi}{5}$ D) $\frac{205\pi}{9}$

$P = \frac{abc}{4R}$

VIDEO SOLUTION

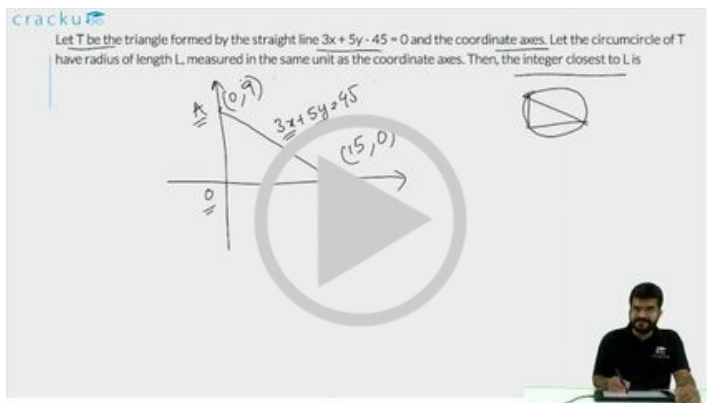
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Question 4

Let T be the triangle formed by the straight line $3x + 5y - 45 = 0$ and the coordinate axes. Let the circumcircle of T have radius of length L , measured in the same unit as the coordinate axes. Then, the integer closest to L is

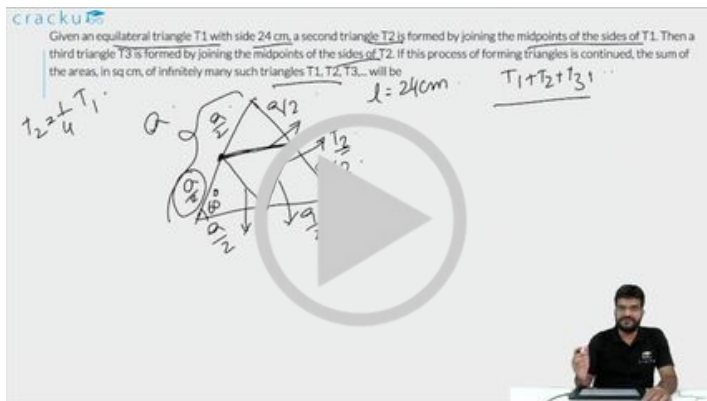


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Question 5

Given an equilateral triangle T_1 with side 24 cm, a second triangle T_2 is formed by joining the midpoints of the sides of T_1 . Then a third triangle T_3 is formed by joining the midpoints of the sides of T_2 . If this process of forming triangles is continued, the sum of the areas, in sq cm, of infinitely many such triangles $T_1, T_2, T_3, \dots$ will be

- A $188\sqrt{3}$
- B $248\sqrt{3}$
- C $164\sqrt{3}$
- D $192\sqrt{3}$



VIDEO SOLUTION

Question 6

A triangle ABC has area 32 sq units and its side BC , of length 8 units, lies on the line $x = 4$. Then the shortest possible distance between A and the point $(0,0)$ is

- A 8 units
- B 4 units
- C $2\sqrt{2}$ units
- D $4\sqrt{2}$ units

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A triangle ABC has area 32 sq units and its side BC, of length 8 units, lies on the line $x = 4$. Then the shortest possible distance between A and the point (0,0) is

Handwritten calculations:

$$\frac{1}{2} \times 8 \times h = 32$$

$$4h = 32$$

$$h = 8 \text{ units}$$

8 units

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Question 7

The area of the closed region bounded by the equation

$|x| + |y| = 2$ in the two-dimensional plane is

- A 4π sq. units
- B 4 sq. units
- C 8 sq. units
- D 2π sq. units

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The area of the closed region bounded by the equation $|x| + |y| = 2$ in the two-dimensional plane is

Handwritten equations:

$$x+y=2; x-y=2; -x+y=2; -x-y=2$$

8 sq. units

VIDEO SOLUTION

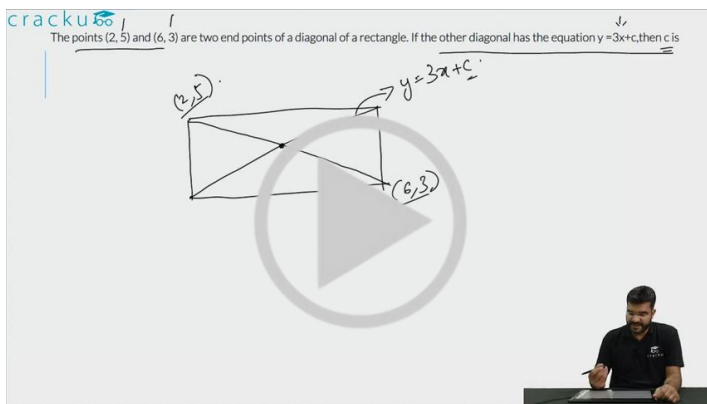
Question 8

The points (2, 5) and (6, 3) are two end points of a diagonal of a rectangle. If the other diagonal has the equation $y = 3x + c$, then c is

- A -5
- B -6

C -7

D -8



VIDEO SOLUTION

Question 9

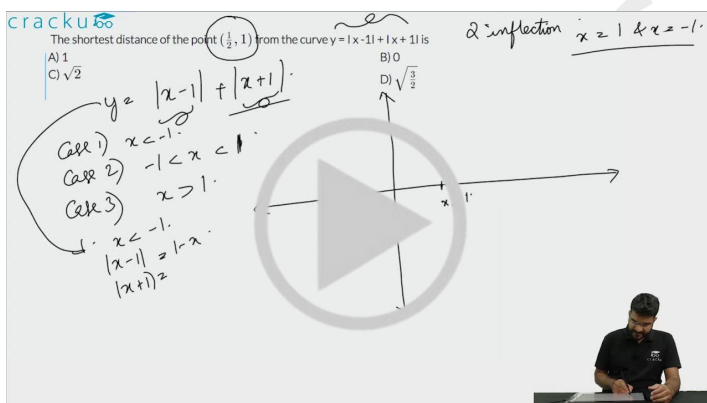
The shortest distance of the point $(\frac{1}{2}, 1)$ from the curve $y = |x - 1| + |x + 1|$ is

A 1

B 0

C $\sqrt{2}$

D $\sqrt{\frac{3}{2}}$



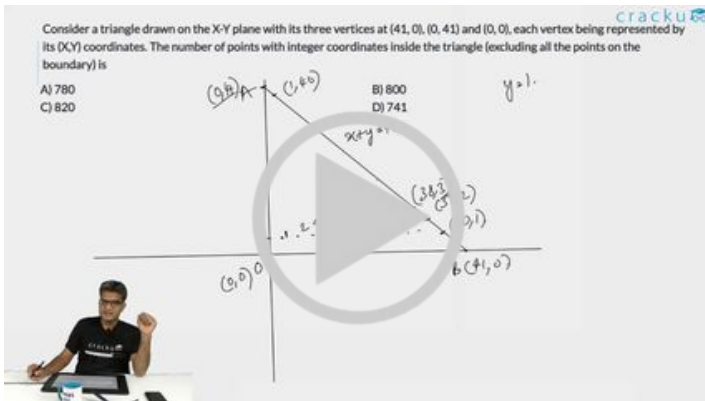
VIDEO SOLUTION



Question 10

Consider a triangle drawn on the X-Y plane with its three vertices at (41, 0), (0, 41) and (0, 0), each vertex being represented by its (X,Y) coordinates. The number of points with integer coordinates inside the triangle (excluding all the points on the boundary) is

- A 780
- B 800
- C 820
- D 741

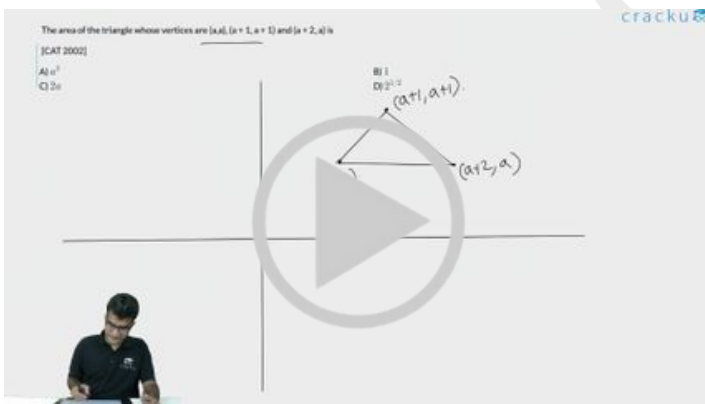


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Question 11

The area of the triangle whose vertices are (a,a) , $(a+1, a+1)$ and $(a+2, a)$ is

- A a^3
- B 1
- C $2a$
- D $2^{1/2}$



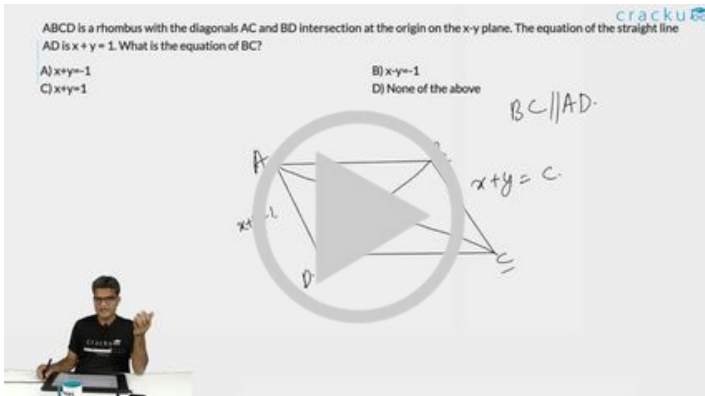
VIDEO SOLUTION

Question 12

ABCD is a rhombus with the diagonals AC and BD intersection at the origin on the x-y plane. The equation of the straight line AD is $x + y = 1$. What is the equation of BC?

- A $x+y=-1$

- B $x-y=-1$
- C $x+y=1$
- D None of the above



VIDEO SOLUTION



CAT Syllabus PDF



Question 13

The points of intersection of three lines $2x + 3y - 5 = 0$, $5x - 7y + 2 = 0$ and $9x - 5y - 4 = 0$

- A form a triangle
- B are on lines perpendicular to each other
- C are on lines parallel to each other
- D are coincident

VIEW SOLUTION

Question 14

What is the distance between the points A(3, 8) and B(-2,-7)?

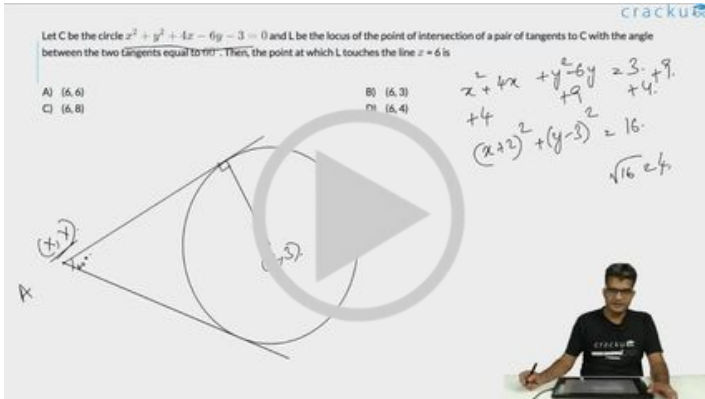
- A $5\sqrt{2}$
- B 5
- C $5\sqrt{10}$
- D $10\sqrt{2}$

VIEW SOLUTION

Question 15

Let C be the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ and L be the locus of the point of intersection of a pair of tangents to C with the angle between the two tangents equal to 60° . Then, the point at which L touches the line $x = 6$ is

- A (6, 6)
- B (6, 3)
- C (6, 8)
- D (6, 4)

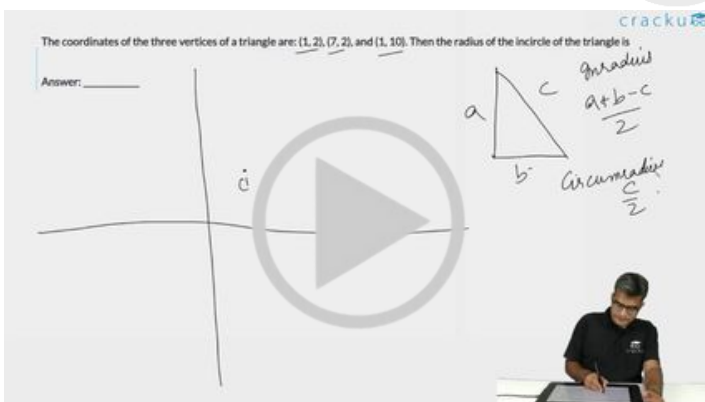


VIDEO SOLUTION



Question 16

The coordinates of the three vertices of a triangle are: (1, 2), (7, 2), and (1, 10). Then the radius of the incircle of the triangle is



VIDEO SOLUTION

Question 17

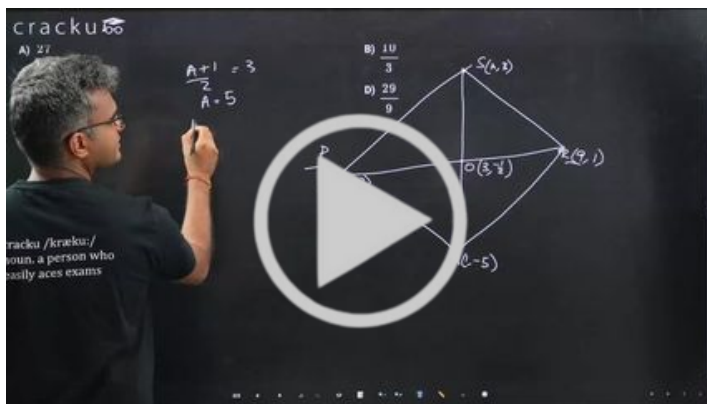
The (x, y) coordinates of vertices P, Q and R of a parallelogram PQRS are (-3, -2), (1, -5) and (9, 1), respectively. If the diagonal SQ intersects the x-axis at (a, 0), then the value of a is

A $\frac{27}{7}$

B $\frac{10}{3}$

C $\frac{13}{4}$

D $\frac{29}{9}$



 VIDEO SOLUTION

Answers

1.D	2.D	3.D	4.9	5.D	6.B	7.C	8.D
9.A	10.A	11.B	12.A	13.D	14.C	15.B	16.2
17.D							

Explanations

1. D

In a parallelogram, two diagonals of parallelogram bisect each other, which concludes that mid-point of both diagonals are the same.

$$\text{Midpoint of AC} = \left(\frac{1-2}{2}, \frac{1+8}{2} \right)$$

Let the coordinates of vertex D be (x,y)

$$\left(\frac{x+3}{2}, \frac{y+4}{2} \right) = \left(\frac{1-2}{2}, \frac{1+8}{2} \right)$$

$$x = -4 \text{ and } y = 5$$

The answer is option D.

 VIDEO SOLUTION

2. D

The midpoints of two diagonals of a parallelogram are the same

Hence the midpoint of (2,1) and (-3,-4) lie on $x + 9y + c = 0$

$$\text{midpoint of (2,1) and (-3,-4)} = \left(\frac{2-3}{2}, \frac{1-4}{2} \right) = (-1/2, -3/2)$$

Keeping this coordinates in the above line equation, we get $c = 14$

 VIDEO SOLUTION

3. D

$$\text{Equation of circle } x^2 + y^2 + 2gx + 2fy + c = 0$$

It passes through (0,0), (4,0) and (3,9). Substitute each point in the above equation:

$$\Rightarrow \text{On substituting the value (0,0) in the above equation, we obtain: } c = 0$$

$$\Rightarrow \text{On substituting the value (4,0) in the above equation, we obtain: } 16 + 0 + 8g + 0 = 0; g = -2$$

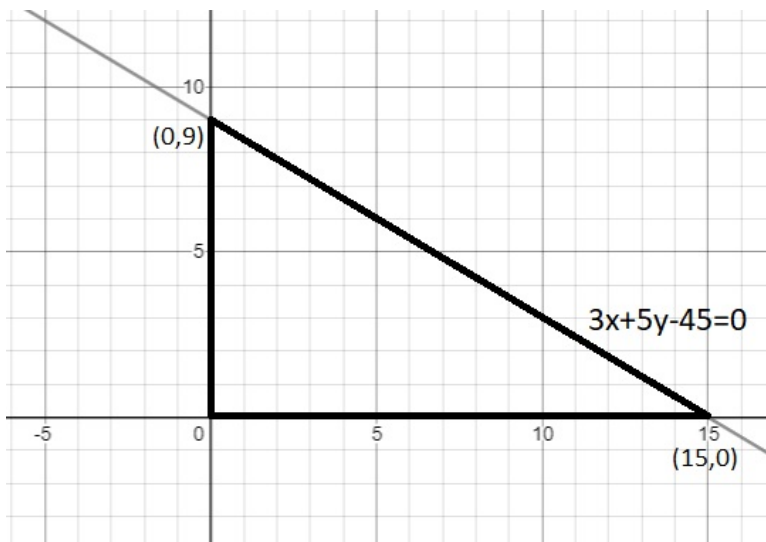
$$\Rightarrow \text{On substituting the value (3,9) in the above equation, we obtain: } 9 + 81 - 12 + 18f = 0; f = -13/3$$

$$\text{Radius of the circle } r = \sqrt{g^2 + f^2 - c} \Rightarrow r^2 = \frac{205}{9}$$

$$\text{Therefore, Area} = \pi r^2 = \frac{205\pi}{9}$$

 VIDEO SOLUTION

4. 9



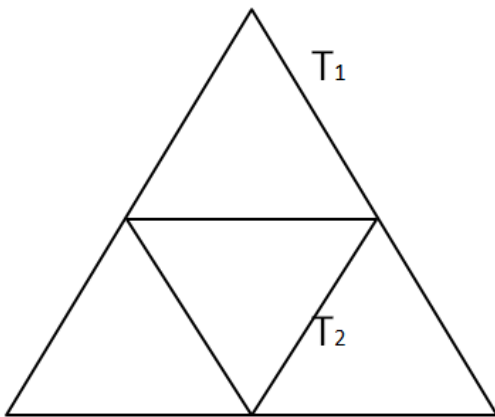
In any right triangle, the circumradius is half of the hypotenuse. Here, $L = \frac{1}{2} \times \text{the length of the hypotenuse} = \frac{1}{2}(\sqrt{15^2 + 9^2}) = \frac{1}{2} \times \sqrt{306} = \frac{1}{2} \times 17.49 = 8.74$

Hence, the integer close to $L = 9$

[VIDEO SOLUTION](#)

5. D

We can see that T_2 is formed by using the mid points of T_1 . Hence, we can say that area of triangle of T_2 will be $(1/4)$ th of the area of triangle T_1 .



$$\text{Area of triangle } T_1 = \frac{\sqrt{3}}{4} \times (24)^2 = 144\sqrt{3} \text{ sq. cm}$$

$$\text{Area of triangle } T_2 = \frac{144\sqrt{3}}{4} = 36\sqrt{3} \text{ sq. cm}$$

$$\text{Sum of the area of all triangles} = T_1 + T_2 + T_3 + \dots$$

$$\Rightarrow T_1 + T_1/4 + T_1/4^2 + \dots$$

$$\Rightarrow \frac{T_1}{1 - 0.25}$$

$$\Rightarrow \frac{4}{3} \times T_1$$

$$\Rightarrow \frac{4}{3} \times 144\sqrt{3}$$

$$\Rightarrow 192\sqrt{3}$$

Hence, option D is the correct answer.

 VIDEO SOLUTION

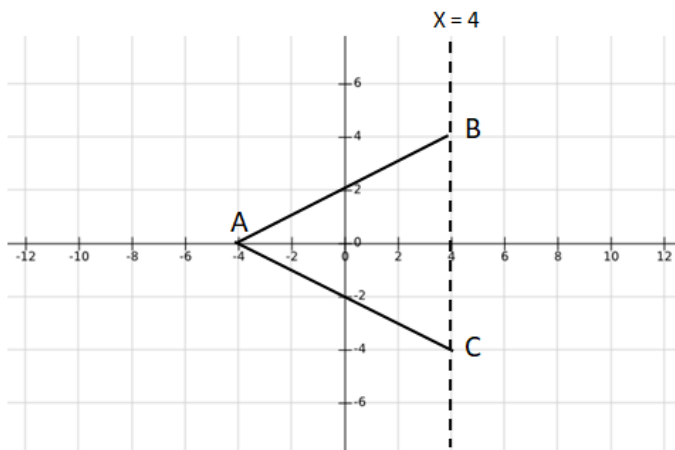


6. B

We know that area of the triangle = 32 sq. units, $BC = 8$ units

Therefore, the height of the perpendicular drawn from point A to $BC = 2 \times 32 / 8 = 8$ units.

Let us draw a possible diagram of the given triangle.



We can see that if A coincide with $(-4, 0)$ then the distance between A and $(0, 0) = 4$ units.

If we move the triangle up or down keeping the base BC on $x = 4$, then point A will move away from origin as vertical distance will come into factor whereas horizontal distance will remain as 4 units.

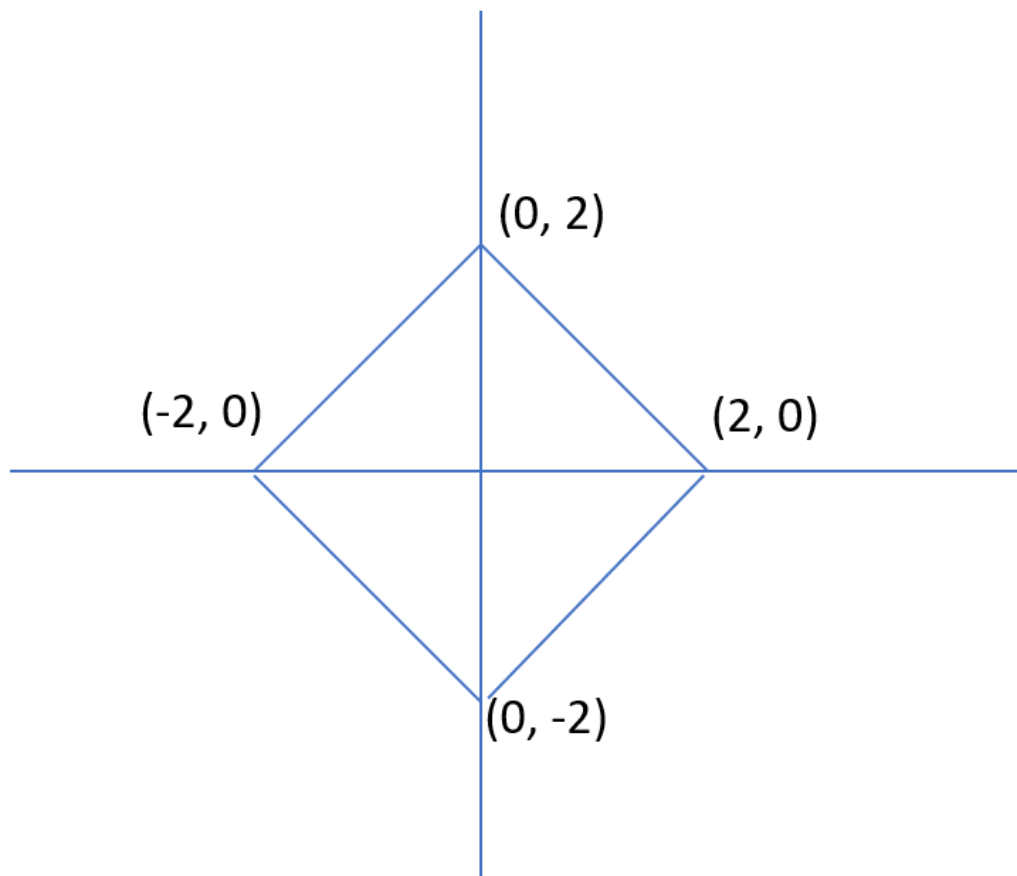
Hence, we can say that minimum distance between A and origin $(0, 0) = 4$ units.

 VIDEO SOLUTION

7. C

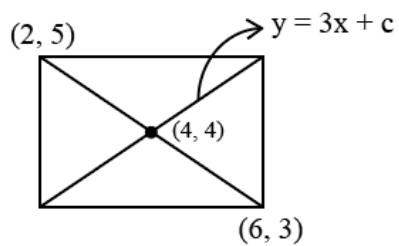
The following equation will form a square of side $2\sqrt{2}$.

The area of the square = $(2\sqrt{2})^2 = 8$ units.



VIDEO SOLUTION

8. D



The midpoint of one diagonal lies on the other diagonal.

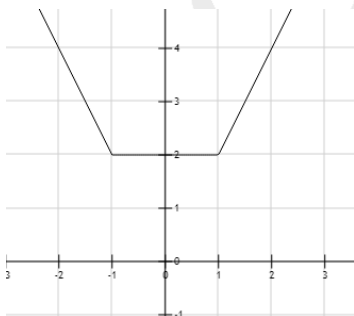
Midpoint is $((2+6)/2, (5+3)/2) = (4, 4)$

Hence $4 = 3 * 4 + c \Rightarrow c = -8$

VIDEO SOLUTION

9. A

The graph of the given function is as shown below.



We can see that the shortest distance of the point $(1/2, 1)$ will be 1 unit.

[VIDEO SOLUTION](#)

10. A

The number of points on $x = 1$ is 39. The number of points on $x = 2$ is 38 and so on till $x = 39$, which has one point.

So, the total is $1+2+3+\dots+39 = \frac{39 \times 40}{2} = 780$.

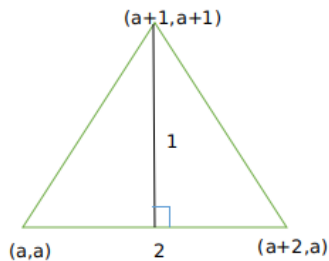
[VIDEO SOLUTION](#)

CAT Percentile Predictor



11. B

The triangle we have is :



The length of three sides is $\sqrt{2}$, $\sqrt{2}$ and 2.

This is a right-angled triangle.

Hence, its area equals $\frac{1}{2} \times \sqrt{2} \times \sqrt{2} = 1$

So, the correct answer is b)

Alternate Approach :

Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

So we get $\frac{1}{2} \times 2 \times 1 = 1$ square units

[VIDEO SOLUTION](#)

12. A

The line should be parallel to AD and should be of equal distance from the origin in the third quadrant. The equation $x+y = -1$ satisfies all these conditions.

[VIDEO SOLUTION](#)

13. D

For points to be coincident, value of determinant should not be equal zero, so that they have a unique value of system.

Here value of determinant is not equal to zero, simultaneously not any two lines are parallel or perpendicular.

So system has a unique value

Hence points are coincident.

[VIEW SOLUTION](#)

14.C

Distance between two points = $\sqrt{(3 - (-2))^2 + (8 - (-7))^2} = 5\sqrt{10}$

[VIEW SOLUTION](#)

15.B

Given equation of circle = $x^2 + y^2 + 4x - 6y - 3 = 0$

Center of the circle is (-2,3) and radius of the circle = $\sqrt{g^2 + f^2 - c} = \sqrt{4 + 9 + 3} = 4$

Let us assume the point of the intersection of the tangents is (h,k)

The angle made by the line joining (h,k) to the centre makes an angle of 30 degrees with the tangent, and $\sin(30)$ will be the ratio of the radius and the distance between the center and (h,k)

$$\Rightarrow \sin(30) = \frac{4}{\sqrt{(h+2)^2 + (k-3)^2}}$$

Squaring on both sides:

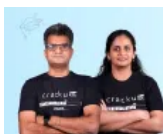
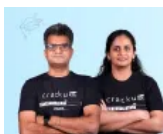
$$\frac{1}{4} = \frac{16}{(h+2)^2 + (k-3)^2}$$

$$\Rightarrow (h+2)^2 + (k-3)^2 = 64$$

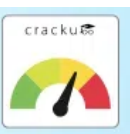
When $x = 6 \Rightarrow h = 6 \Rightarrow 64 + (k-3)^2 = 64 \Rightarrow k = 3$.

$\Rightarrow$ required point is (6,3)

[VIDEO SOLUTION](#)

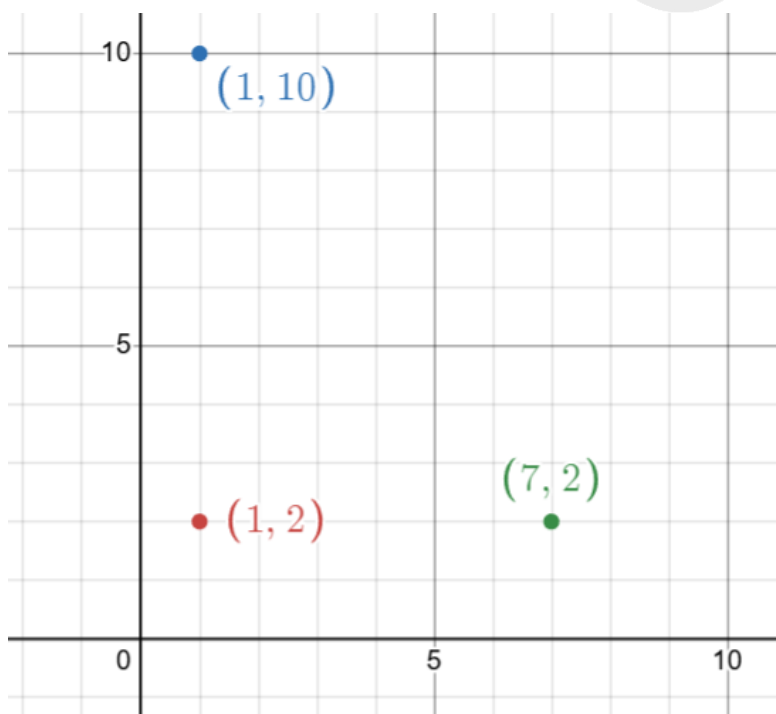


CAT Score Calculator



16.2

Upon drawing a rough sketch of the coordinates given, we realise that this is a right-angled triangle.



The three side lengths are 6, 8 and 10 units

The inradius of a circle can be calculated using the formula: $rs = \text{Area}$, where s is the semi-perimeter of the triangle

The Area would be $\frac{1}{2} \times 6 \times 8 = 24$
and the semi-perimeter would be $\frac{10+6+8}{2} = 12$

Giving the inradius to be $\frac{24}{12} = 2$ units

Therefore, 2 is the correct answer.

 VIDEO SOLUTION

17. D

Given $(P(-3,-2))$, $(Q(1,-5))$, and $(R(9,1))$.

For parallelogram (PQRS), $S = P + R - Q = (-3,-2) + (9,1) - (1,-5) = (5,4)$

Diagonal SQ passes through $S(5,4)$ and $Q(1,-5)$.

$$\text{Slope } m = \frac{-5 - 4}{1 - 5} = \frac{9}{4}$$

$$\text{Equation of SQ is } y - 4 = \frac{9}{4}(x - 5)$$

$$\text{At the x-axis, } y = 0: -4 = \frac{9}{4}(a - 5)$$

$$-16 = 9(a - 5)$$

$$9a = 29$$

$$a = \frac{29}{9}$$

$$\text{So, } a = \frac{29}{9}$$

 VIDEO SOLUTION